JEE Main 2027 — Model Question Paper 3

Duration: 3 hours · Maximum marks: 300 · 75 questions (25 each in Physics, Chemistry, Mathematics)

1. Section A of each subject: 20 multiple-choice questions, one correct option. +4 for correct, −1 for wrong, 0 if unattempted.
2. Section B of each subject: 5 numerical-value questions; enter the exact integer answer. +4 for correct, 0 otherwise (no negative marking).
3. Take g = 10 m/s² unless stated otherwise.
4. Answer key is at the end of the paper.

Physics

Section A — Multiple Choice (single correct)

1. The dimensional formula of Planck's constant is

(1) $[ML^2T^{-2}]$ **(2)** $[ML^2T^{-1}]$
(3) $[MLT^{-1}]$ **(4)** $[ML^{-1}T^{-1}]$

2. Rain falls vertically at 10 m/s and a person walks at 10 m/s. To keep dry, the umbrella should be tilted forward from the vertical by

(1) 45° **(2)** 30°
(3) 60° **(4)** 0°

3. A 0.1 kg ball hits a wall normally at 20 m/s and rebounds with the same speed. The impulse on the ball has magnitude

(1) 2 N s **(2)** 0
(3) 4 N s **(4)** 20 N s

4. A pump lifts 100 kg of water through 10 m in 10 s (g = 10 m/s²). Its minimum power is

(1) 100 W **(2)** 10 kW
(3) 500 W **(4)** 1 kW

5. The centre of mass of a uniform semicircular ring of radius R lies from its centre at

(1) $4R/3\pi$ **(2)** $2R/\pi$
(3) $R/2$ **(4)** R/π

6. A force of 10 N acts perpendicular to a spanner at 0.5 m from the bolt. The torque is

(1) 5 N m **(2)** 20 N m
(3) 10 N m **(4)** 2.5 N m

7. If the orbital radius of a planet is made 4 times, its period becomes

(1) 4 times **(2)** 16 times
(3) 8 times **(4)** 2 times

8. The excess pressure inside a soap bubble of radius r and surface tension T is

(1) $2T/r$ **(2)** T/r
(3) $8T/r$ **(4)** $4T/r$

9. Two rods of the same material have lengths in the ratio 1 : 2 and areas 2 : 1. For the same temperature difference, the ratio of heat flow rates is

(1) 1 : 4 **(2)** 4 : 1
(3) 1 : 1 **(4)** 2 : 1

10. If the absolute temperature of a black body is doubled, the wavelength of maximum emission

(1) halves **(2)** doubles

(3) becomes ¼ **(4)** is unchanged

11. If the absolute temperature of a black body is doubled, its emitted power becomes

(1) 2 times **(2)** 4 times

(3) 16 times **(4)** 8 times

12. In SHM of amplitude A, kinetic and potential energies are equal at displacement

(1) A/2 **(2)** A

(3) A/4 **(4)** A/√2

13. For the wave $y = 0.05 \sin 2\pi(10t - x/2)$ in SI units, the wave speed is

(1) 5 m/s **(2)** 20 m/s

(3) 10 m/s **(4)** 0.2 m/s

14. An electric dipole is placed in a uniform electric field. It experiences

(1) zero net force, possibly non-zero torque **(2)** a net force and no torque

(3) a net force and torque **(4)** neither force nor torque in any orientation

15. Capacitors of 2 μF and 3 μF are connected in series. The equivalent capacitance is

(1) 5 μF **(2)** 6 μF

(3) 1.2 μF **(4)** 2.5 μF

16. If the current through a given wire is doubled, the drift velocity of electrons

(1) halves **(2)** is unchanged

(3) becomes four times **(4)** doubles

17. Two long parallel wires carry currents in the same direction. They

(1) repel **(2)** attract

(3) exert no force **(4)** experience only torque

18. Energy stored in an inductor of 2 H carrying 3 A is

(1) 9 J **(2)** 6 J

(3) 18 J **(4)** 3 J

19. The Brewster angle for a medium of refractive index √3 is

(1) 30° **(2)** 45°

(3) 60° **(4)** 90°

20. In an n-type semiconductor, the majority charge carriers are

(1) holes **(2)** protons

(3) ions **(4)** electrons

Section B — Numerical Value

21. A ball is thrown vertically up at 30 m/s ($g = 10\ m/s^2$). The time (in s) after which it returns to the thrower is

Answer: _____

22. The equivalent resistance (in Ω) of 4 Ω and 12 Ω in parallel is

Answer: _____

23. The ratio of the radius of a nucleus of mass number 64 to that of mass number 1 is

Answer: _____

24. The power (in dioptre) of a convex lens of focal length 25 cm is

Answer: _____

25. A string fixed at both ends vibrates in its third harmonic. The number of nodes, including the ends, is

Answer: _____

Chemistry

Section A — Multiple Choice (single correct)

26. Which set of quantum numbers is not allowed?

(1) n = 2, l = 1, m = 0 **(2)** n = 2, l = 2, m = 1

(3) n = 3, l = 2, m = −2 **(4)** n = 1, l = 0, m = 0

27. Which species is isoelectronic with Ne?

(1) Na^+ **(2)** K^+

(3) Cl^- **(4)** Ar

28. The largest ion among the following is

(1) F^- **(2)** Na^+

(3) N^{3-} **(4)** O^{2-}

29. Hydrogen bonding is strongest in

(1) H_2O **(2)** NH_3

(3) H_2S **(4)** HF

30. Which molecule is paramagnetic?

(1) N_2 **(2)** O_2

(3) F_2 **(4)** H_2

31. The ratio of rates of effusion of H_2 to O_2 under identical conditions is

(1) 4 : 1 **(2)** 1 : 4

(3) 16 : 1 **(4)** 2 : 1

32. Kohlrausch's law is concerned with

(1) osmotic pressure **(2)** vapour pressure

(3) limiting molar conductivity **(4)** heat of reaction

33. Temporary hardness of water is due to

(1) chlorides of Ca and Mg **(2)** sulphates of Ca and Mg

(3) NaCl **(4)** bicarbonates of Ca and Mg

34. For a spontaneous process at constant T and P

(1) $\Delta G > 0$ **(2)** $\Delta G < 0$

(3) $\Delta S < 0$ **(4)** $\Delta H > 0$ always

35. If K is the equilibrium constant of a reaction, that of its reverse reaction is

(1) 1/K **(2)** K

(3) −K **(4)** K^2

36. An example of an ambident nucleophile is

(1) NH_3 **(2)** H_2O

(3) CN^- **(4)** Cl^-

37. A Grignard reagent reacts with CO_2 followed by hydrolysis to give

(1) an alcohol **(2)** an aldehyde

(3) a ketone **(4)** a carboxylic acid

38. Benzene with CH_3Cl in presence of anhydrous $AlCl_3$ gives

(1) chlorobenzene **(2)** toluene

(3) benzyl chloride **(4)** xylene only

39. Reimer–Tiemann reaction of phenol gives

(1) salicylaldehyde **(2)** benzaldehyde

(3) salicylic acid only **(4)** picric acid

40. The carbylamine test is given by

(1) secondary amines **(2)** tertiary amines

(3) primary amines **(4)** amides

41. The most stable carbocation is

(1) CH_3^+ **(2)** $CH_3CH_2^+$

(3) $(CH_3)_2CH^+$ **(4)** $(CH_3)_3C^+$

42. The intense colour of $KMnO_4$ is due to

(1) d–d transitions **(2)** charge transfer

(3) unpaired electrons **(4)** f–f transitions

43. The CFSE of a low-spin d^6 octahedral complex (ignoring pairing energy) is

(1) $-2.4\ \Delta_o$ **(2)** $-0.4\ \Delta_o$

(3) $-1.6\ \Delta_o$ **(4)** 0

44. The most acidic oxide is

(1) Na_2O **(2)** MgO

(3) Cl_2O_7 **(4)** Al_2O_3

45. Deficiency of vitamin C causes

(1) rickets **(2)** night blindness

(3) beriberi **(4)** scurvy

Section B — Numerical Value

46. The number of radial nodes in a 3s orbital is

Answer: _____

47. The volume (in mL) of 0.1 mol of an ideal gas at STP (molar volume 22.4 L) is

Answer: _____

48. The mass (in g) of NaOH needed to neutralise 1 L of 0.1 M HCl is

Answer: _____

49. The number of chiral carbon atoms in open-chain D-glucose is

Answer: _____

50. The coordination number of Fe in $[Fe(C_2O_4)_3]^{3-}$ is

Answer: _____

Mathematics

Section A — Multiple Choice (single correct)

51. The number of subsets of {1, 2, 3} is

(1) 6 **(2)** 8
(3) 3 **(4)** 9

52. The domain of $f(x) = \sqrt{(x - 2)}$ is

(1) $[2, \infty)$ **(2)** $(2, \infty)$
(3) $(-\infty, 2]$ **(4)** $\mathbb{R}$

53. $\tan 75^\circ$ equals

(1) $\sqrt{3}$ **(2)** $2 - \sqrt{3}$
(3) $2 + \sqrt{3}$ **(4)** $1/\sqrt{3}$

54. If $\cos\theta = 3/5$ and θ is acute, $\tan\theta$ is

(1) 3/4 **(2)** 4/5
(3) 5/3 **(4)** 4/3

55. $(1 + i)^8$ equals

(1) 8 **(2)** 16
(3) 16i **(4)** -16

56. If ω is a non-real cube root of unity, $\omega^{100} + \omega^{200}$ equals

(1) -1 **(2)** 1
(3) 0 **(4)** 2

57. The number of ways of choosing 3 objects out of 7 is

(1) 210 **(2)** 21
(3) 35 **(4)** 70

58. The AM of two positive numbers is 10 and their GM is 8. The numbers are

(1) 12, 8 **(2)** 18, 2
(3) 10, 10 **(4)** 16, 4

59. The slope of a line perpendicular to $2x + 3y = 5$ is

(1) $-2/3$ **(2)** 3/2
(3) 2/3 **(4)** $-3/2$

60. The radius of the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ is

(1) 5 **(2)** $\sqrt{12}$
(3) 25 **(4)** $\sqrt{13}$

61. $\lim_{x\to\infty} (1 + 1/x)^x$ equals

(1) 1 **(2)** 0
(3) e **(4)** ∞

62. $\frac{d}{dx}(\sin^{-1} x)$ equals

(1) $1/(1 + x^2)$ **(2)** $-1/\sqrt{(1 - x^2)}$
(3) $\sqrt{(1 - x^2)}$ **(4)** $1/\sqrt{(1 - x^2)}$

63. $f(x) = x^3 - 3x$ has a local minimum at

(1) $x = -1$ **(2)** $x = 1$
(3) $x = 0$ **(4)** $x = 3$

64. $\int_0^{\pi} \sin x\, dx$ equals

(1) 2 **(2)** 0

(3) 1 **(4)** π

65. $\int dx/(1 + x^2)$ equals

(1) $\ln(1 + x^2) + C$ **(2)** $\sin^{-1} x + C$

(3) $\tan^{-1} x + C$ **(4)** $\cot x + C$

66. The integrating factor of $dy/dx + y/x = x$ is

(1) $1/x$ **(2)** e^x

(3) $\ln x$ **(4)** x

67. For $a = 2\hat{i} + \hat{j} - \hat{k}$ and $b = \hat{i} - \hat{j} + \hat{k}$, the vectors are

(1) parallel **(2)** perpendicular

(3) equal **(4)** collinear and opposite

68. A fair coin is tossed 4 times. The probability of exactly 2 heads is

(1) 3/8 **(2)** 1/2

(3) 1/4 **(4)** 5/16

69. The mean of 5 observations is 10. If one observation 14 is replaced by 4, the new mean is

(1) 6 **(2)** 10

(3) 8 **(4)** 9

70. If $\tan A = 1/2$ and $\tan B = 1/3$ (A, B acute), then $A + B$ equals

(1) $\pi/3$ **(2)** $\pi/6$

(3) $\pi/2$ **(4)** $\pi/4$

Section B — Numerical Value

71. The number of integers x satisfying $x^2 - 5x + 4 < 0$ is

Answer: _____

72. The sum of the coefficients in the expansion of $(2x - 1)^7$ is

Answer: _____

73. If $|a| = 3$, $|b| = 4$ and $a \perp b$, then $|a + b|$ is

Answer: _____

74. The value of ${}^5C_0 + {}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5$ is

Answer: _____

75. The number of functions from {1, 2, 3} to {a, b} is

Answer: _____

Answer Key

Physics

1. (2)	**2.** (1)	**3.** (3)	**4.** (4)	**5.** (2)	**6.** (1)
7. (3)	**8.** (4)	**9.** (2)	**10.** (1)	**11.** (3)	**12.** (4)
13. (2)	**14.** (1)	**15.** (3)	**16.** (4)	**17.** (2)	**18.** (1)
19. (3)	**20.** (4)	**21.** 6	**22.** 3	**23.** 4	**24.** 4
25. 4					

Chemistry

26. (2)	**27.** (1)	**28.** (3)	**29.** (4)	**30.** (2)	**31.** (1)
32. (3)	**33.** (4)	**34.** (2)	**35.** (1)	**36.** (3)	**37.** (4)
38. (2)	**39.** (1)	**40.** (3)	**41.** (4)	**42.** (2)	**43.** (1)
44. (3)	**45.** (4)	**46.** 2	**47.** 2240	**48.** 4	**49.** 4
50. 6					

Mathematics

51. (2)	**52.** (1)	**53.** (3)	**54.** (4)	**55.** (2)	**56.** (1)
57. (3)	**58.** (4)	**59.** (2)	**60.** (1)	**61.** (3)	**62.** (4)
63. (2)	**64.** (1)	**65.** (3)	**66.** (4)	**67.** (2)	**68.** (1)
69. (3)	**70.** (4)	**71.** 2	**72.** 1	**73.** 5	**74.** 32
75. 8					

Original model paper prepared for practice, following the latest announced exam pattern. Not an official NTA paper.